IT World Essentials: Your Digital Entry Point

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no1: **Problem Statement:** A bank wants to calculate simple interest for customers. Write a C program that takes principal, rate, and time as input and calculates the interest using a function.

Solution:
```
#include <stdio.h>

float calculateSI(float p, float r, float t) {

    return (p * r * t) / 100;

}

int main() {

    float p, r, t;

    printf("Enter Principal, Rate, Time: ");

    scanf("%f %f %f", &p, &r, &t);

    printf("Simple Interest = %.2f\n", calculateSI(p, r, t));

    return 0;

}
```

Description: In banking systems, interest calculation is a common operation. This program simulates a basic banking feature where the user enters the principal amount, rate of interest, and time period.

- **Formula Used:** {Simple Interest} = (P \times R \times T) / 100
- **Input:** Principal: 1000, Rate: 5, Time: 2
- **Output:** Simple Interest = 100.00

Q.no2: **Problem Statement:** A student management system needs to calculate total and average marks of 5 subjects.

Solution:
```
#include <stdio.h>

float getTotal(float m[]) {
```

```
    float s = 0;
    for(int i=0; i<5; i++) s += m[i];
    return s;
}
int main() {
    float marks[5];
    printf("Enter marks for 5 subjects: ");
    for(int i=0; i<5; i++) scanf("%f", &marks[i]);
    float total = getTotal(marks);
    printf("Total = %.0f\nPercentage = %.2f\n", total, total/5);
    return 0;
}
```

Description: The system takes marks of 5 subjects as input and calculates total and percentage using functions.

- **Input:** Marks: 80 70 90 85 75
- **Output:** Total = 400, Percentage = 80.00

Q.no3: **Problem Statement:** An ATM machine needs to check whether the entered PIN is correct.

Solution:
```
#include <stdio.h>
void checkPIN(int pin) {
    if(pin == 1234) printf("Access Granted\n");
    else printf("Invalid PIN\n");
}
int main() {
    int input;
    printf("Enter PIN: ");
    scanf("%d", &input);
```

```
    checkPIN(input);
    return 0;
}
```

Description: In an ATM, before doing any transaction, the user must enter a correct PIN. If the PIN is correct (e.g., 1234), the user can access the system; otherwise, access is denied.

- **Input:** Enter PIN: 1234
- **Output:** Access Granted

Q.no4: **Problem Statement:** A company wants to check whether an employee ID number is a palindrome.

Solution:
```
#include <stdio.h>
int isPalindrome(int id) {
    int rev = 0, temp = id;
    while(temp != 0) {
        rev = rev * 10 + (temp % 10);
        temp /= 10;
    }
    return (rev == id);
}
int main() {
    int id;
    printf("Enter ID: ");
    scanf("%d", &id);
    if(isPalindrome(id)) printf("Palindrome\n");
    else printf("Not Palindrome\n");
    return 0;
}
```

Description: A palindrome number remains the same when its digits are reversed (e.g., 121).

- **Input:** Enter ID: 121
- **Output:** Palindrome

.no5: **Problem Statement:** A library system needs to calculate late fine for books returned after the due date.

Solution:
```
#include <stdio.h>

int calculateFine(int days)

{

    if(days > 5) return days * 2;

    return 0;

}

int main()

 {

    int days;

    printf("Enter number of late days: ");

    scanf("%d", &days);

    int fine = calculateFine(days);

    if(fine > 0) printf("Fine = %d\n", fine);

    else printf("No Fine\n");

    return 0;

}
```

Description: If a book is returned late, a fine is charged: if days > 5, the fine is ₹2 per day; otherwise, there is no fine.

- **Input:** Days = 7
- **Output:** Fine = 14